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MORE THAN MEDITATION

Audiomodule

Übersicht Audiomodule

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text

music

nature

sound

Systemsound zum Bluetooth-Click-DeBug

Demo der ausgewählten Stimme (js/main.js)

```
////////////////////////////////////////////////////////////////////////////////
//////////////////////////////// Bluetooth-Headphones-Click-DeBug ////////////

    var isPlayingSilence = false;

    function playSilence() {
        if (isPlayingSilence) {
            return;
        }

        isPlayingSilence = true;

        $('#audioInfinite').trigger("play");
        document.getElementById('audioInfinite').onended = function() {
            isPlayingSilence = false;
            playSilence();
        };
    }
```

noise

```
<!-- ------------------------------------------ -->
<!-- ----------------AUDIO--------------------- -->
<!-- ------------------------------------------ -->
    <div id="audioPlayer" hidden>
        <audio id="textAudio" src="audio/system/platzhalter.mp3" preload="auto">
            <source id="textSource" src="audio/system/platzhalter.mp3" type="audio/mp3">
        </audio>

        <audio id="musicAudio" src="audio/system/platzhalter.mp3" preload="auto">
            <source id="musicSource" src="audio/system/platzhalter.mp3" type="audio/mp3">
        </audio>

        <audio id="natureAudio" src="audio/system/platzhalter.mp3" preload="auto">
            <source id="natureSource" src="audio/system/platzhalter.mp3" type="audio/mp3">
        </audio>

        <audio id="soundAudio" src="audio/system/platzhalter.mp3" preload="auto">
            <source id="soundSource" src="audio/system/platzhalter.mp3" type="audio/mp3">
        </audio>

        <audio id="audioInfinite" src="audio/system/endlessSilence.mp3">
            <source id="sourceInfinite" src="audio/system/endlessSilence.mp3" type="audio/mp3">
        </audio>

        <audio id="audioDemo" src="audio/system/platzhalter.mp3">
            <source id="sourceDemo" src="audio/system/platzhalter.mp3" type="audio/mp3">
        </audio>

    </div>

<!-- ------------------------------------------ -->
```

AWAVE.html

```
        session[phase].noise.audio  = new Audio(src);
        session[phase].noise.source = session[phase].noise.ctx.createMediaElementSource( session[phase].noise.audio );
        session[phase].noise.audio.play();
        session[phase].noise.audio.loop = true;

      // Split Source L/R-Channel
        session[phase].noise.splitter = session[phase].noise.ctx.createChannelSplitter(2);
```

js/generator-frequency-noise.js

Übersicht Audiomodule

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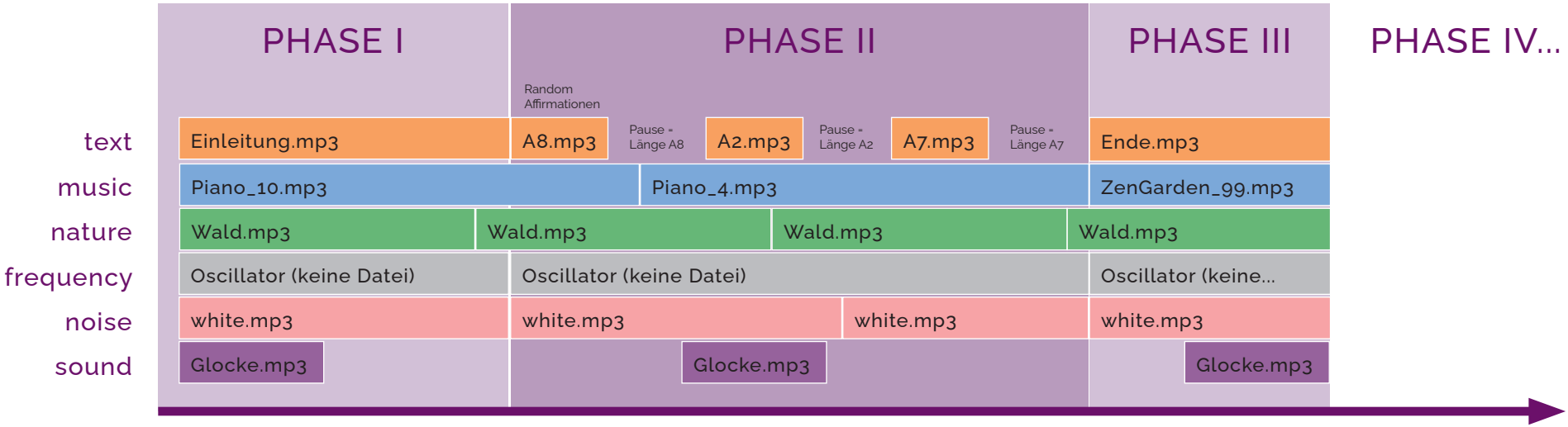
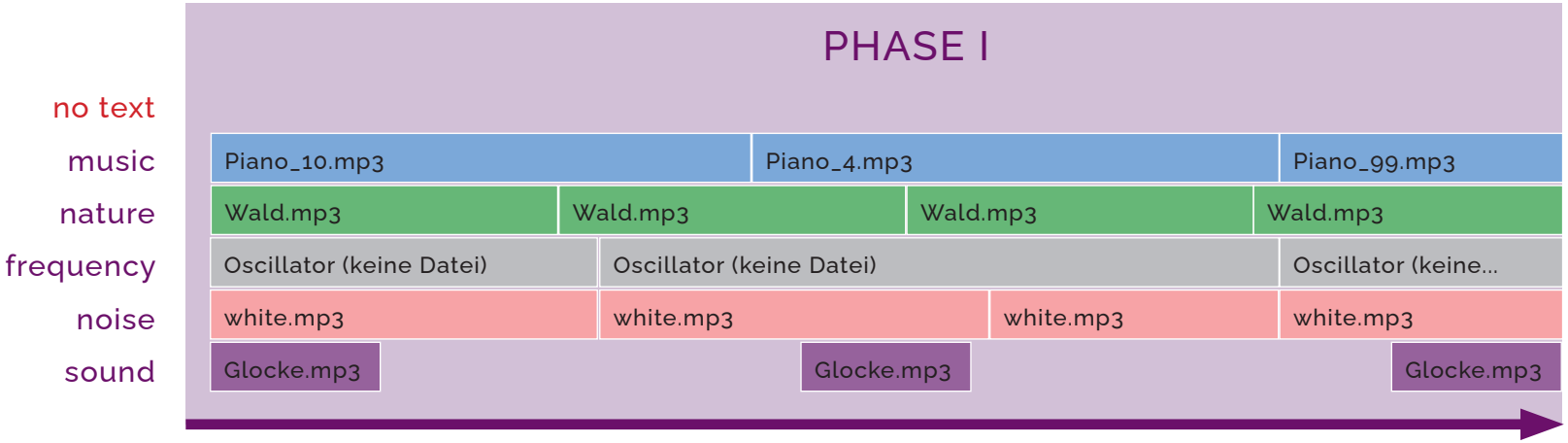
Geführte Meditation:



Klangwelt / Soundscape:



Startpunkte im UI

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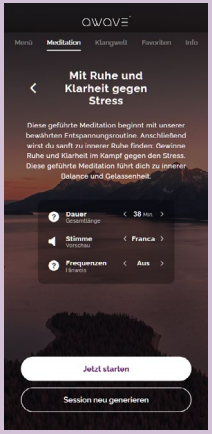
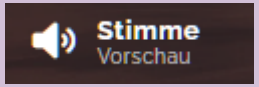
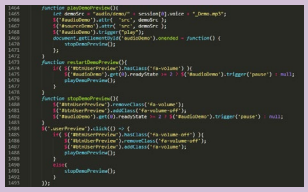
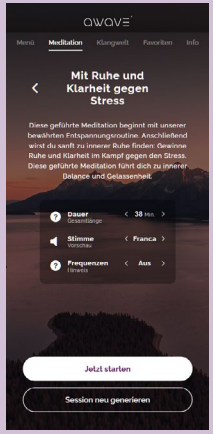
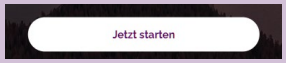




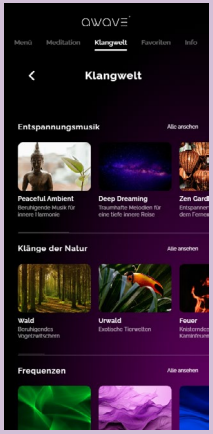


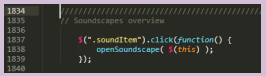
js/main.js 1464

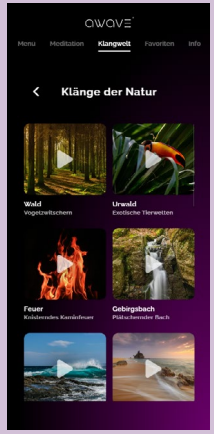




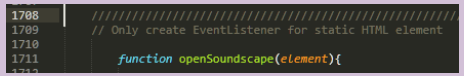
js/main.js 564

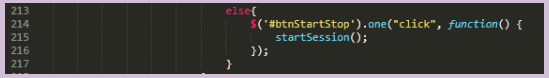




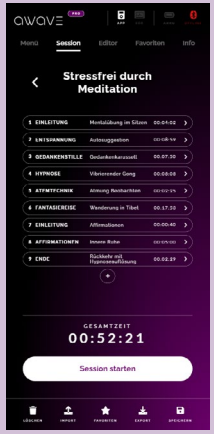


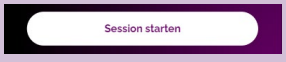
js/content-database.js



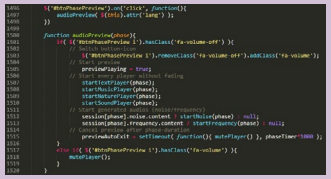


js/editor-live.js 215





js/main.js 561



js/main.js 1507ff

startSession(){ startPhase(1) }

js/main.js 527+555

Start einer Session

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startPhase(phase) in main.js
Steuert zuerst das Verhalten der einzelnen Player über die Phasendauer wie z.B. Fade-In, Fade-Out, etc. und ruft dann die Funktion zur Übergabe der SRC und zum Abspielen auf:
startTextPlayer(phase);
startMusicPlayer(phase);
startNaturePlayer(phase);
startSoundPlayer(phase);

Frequenzen und Rauschen werden über die Web-Audio-API separat behandelt.
Siehe: generator-frequency-noise.js

Frequenzen werden von Oszillatoren in der API generiert
(müssen also nicht bereitgestellt werden).
startFrequency(phase);

Rauschen siehe:
startNoise(phase);

```
527     function startSession() {
554
555         startPhase(1);
556     }
```

```
1006    function startPhase(phase){
1007
1066        ////////////////////////////////////////////////////////////////////////////////
1067
1130            startTextPlayer(phase);
1131
1132
1272            if( !prevMusic  ||  session[phase].music.fadeIn > 1  ||  session[prevPhase].music.fadeOut > 2 ){
1273                startMusicPlayer(phase);
1274            }
1276            // else auto-continue playing and auto gets current phase at end of song
1277
1278
1417            if( !prevNature || session[phase].nature.fadeIn>1 || session[prevPhase].nature.fadeOut>2 ){
1279                startNaturePlayer(phase);
1420            }
1421
1422        ////////////////////////////////////////////////////////////////////////////////
1423        // Generator (Frequency+Noise)
1424            pauseSweep = false;
1425
1426            session[phase].frequency.content ? startFrequency(phase) : null;
1427            session[phase].noise.content ? startNoise(phase) : null;
1428
1429        ////////////////////////////////////////////////////////////////////////////////
1430        // Sound-Interval
1431            $('#soundAudio').prop('volume', session[phase].sound.volume/100);
1432            startSoundPlayer(phase);
1433
1434        ////////////////////////////////////////////////////////////////////////////////
1435        // Refresh Interface at phase start
1436            refreshLiveInterface();
1437    }
```

js/main.js

Content-Änderung im Livebetrieb

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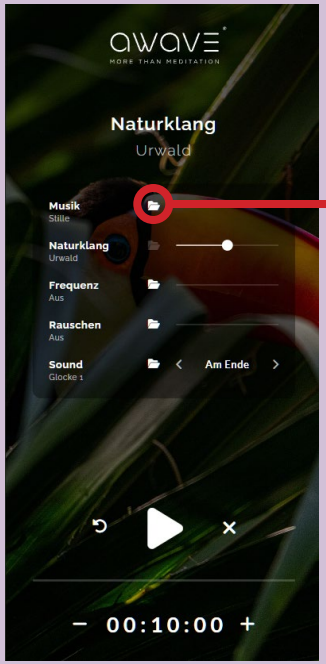

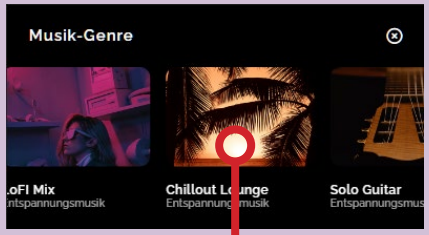


```
44      ////////////////////////////
45      // New Content
46      ////////////////////////////
47          $(".newContentItem").click(function(){
48              navHideContentChange();
49              let newContent = $(this).attr("id").replace("nc_", "");
50              let group = $(this).attr('group');
51
52              // Change session content/config
53                  switch(group){
54
```

js/editor-live.js

```
199                 case "music":
200                 // Multisession + SingleSoundscape
201                     // Multisession
202                     if( session[0].type == "guided" ){
                            startMusicPlayer(session[0].livePhase);
204                         if(playerPaused){ pauseAudio() }
205                     }
206                     // SingleSoundscape
207                     else{
208                         exitSession();
209                         session[0].duration = session[1].duration;
210                         if(!pausedBeforeExit){
211                             setTimeout(function(){startSession()},222);
212                         }
213                         else{
214                             $('#btnStartStop').one("click", function() {
                                    startSession();
216                             });
217                         }
218                     }
219                     refreshLiveInterface();
220                     break;
```

js/editor-live.js

Je nach Case nur Neustart des jeweiligen Players:
startTextPlayer(phase);
startMusicPlayer(phase);
startNaturePlayer(phase);

Automatischer Neustart der kompletten Session:
Case 1: Session enthält nur 1 Phase
Case 2: Timer wurde verändert
Case 3: Neue Frequenz ausgewählt
Case 4: Neues Rauschen ausgewählt
Case 5: Neuer Sound ausgewählt

startSession();

Text-Player

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Aufruf zur Übergabe der src an das Audio-Element mit getAudioSrc().

Bereitstellen der src und Abspielen des Players mit dem Start der jeweiligen Phase:
startTextPlayer(phase)

Case1: textChoice.mix == "one"
Die Länge der Audiodatei bestimmt die Länge der Phase

Case2: textChoice.mix == "loop"
Solange der Timer der Phase noch nicht abgelaufen ist, wird mit onended Zeitversetzt eine weitere Audiodatei mit getAudioSrc() aus dem definierten Content-Pool abgerufen:
session[phase].text.content

```
666  ////////////////////////////
667  // Text
668  ////////////////////////////
669
670      let textDelaySwitch; // Waiting next text-audio ( to clear timeout if needed )
671      let currentDelay;    // Increasing delay value
672      function startTextPlayer(phase){
673          // As a precaution, stop it first if it was running
674              $('#textAudio').trigger('pause');
675
676          // Fill audio-player
677              $('#textAudio').attr(  'src', getAudioSrc( eval(session[phase].text.content) ) );
678              $('#textSource').attr( 'src', getAudioSrc( eval(session[phase].text.content) ) );
679
680          // Play text-audio
681              $('#textAudio').trigger('play');
682
683          // Repeat random if mix=="loop"
684              var textPlayer = document.getElementById('textAudio');
685
686              textPlayer.onended = function() {
687
688                  // Only in preview case
689                      if( !liveMode ){
690                          textDelaySwitch = setTimeout(()=>{ startTextPlayer(phase); },2000);
691                      }
692                  // Only in live-session
693                      else{
694                      // Repeat, increase delay, check remaining phase-time
695                          if( (phaseTimer - ( session[phase].text.delayMin + session[phase].text.delayInc ) ) > 9 ){
696
697                              let textChoice = eval(session[phase].text.content);
698                              if( textChoice.mix == "loop" ){
699
700                                      // Get duration to scale next pause
701                                          if( session[phase].text.delayMin + session[phase].text.delayMax == 0 ){
702                                              currentDelay = $('#textAudio').prop('duration');
703                                              textDelaySwitch = setTimeout(()=>{ startTextPlayer(phase) }, currentDelay*1000 );
704                                          }
705                                          else{
706                                              if( !currentDelay ){
707                                                  currentDelay = session[phase].text.delayMin;
708                                              }
709                                              if( currentDelay + session[phase].text.delayInc  <  session[phase].text.delayMax ){
710                                                  currentDelay += session[phase].text.delayInc;
711                                                  textDelaySwitch = setTimeout(()=>{ startTextPlayer(phase) }, currentDelay*1000 );
712                                              }
713                                              else{
714                                                  textDelaySwitch = setTimeout(()=>{ startTextPlayer(phase); },session[phase].text.delayMax*1000);
715                                              }
716                                          }
717
718
719
720                              } // END_IF loop
721                          } // END-IF phase-time remaining
722                      } // END-ELSE
723              } // END onEnded audio
724      }
```

js/main.js

Music-Player

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Aufruf zur Übergabe der src an das Audio-Element mit getAudioSrc().

Bereitstellen der src und Abspielen des Players mit dem Start der jeweiligen Phase: *startMusicPlayer(phase)*

Case1: thisPhase === nextPhase
Wenn das Musikgenre dieser Phase mit der Nachfolgenden übereinstimmt, wird die Audiodatei am Ende der Phase nicht beendet, sondern fortgesetzt.

Case2: else
Ansonsten wird am Ende des Songs mit onended erneut *startMusicPlayer(phase)* *aufgerufen, sofern der Timer noch nicht abgelaufen ist.*

```
726     ////////////////////////////
727     // Music
728     ////////////////////////////
729
730         let musicDelaySwitch;
731
732         function startMusicPlayer(phase){
733             // As a precaution, stop it first if it was running
734                 $('#musicAudio').trigger('pause');
735
736             // Fill audio-player
737                 $('#musicAudio').attr(  'src', getAudioSrc( eval(session[phase].music.content) ) );
738                 $('#musicSource').attr( 'src', getAudioSrc( eval(session[phase].music.content) ) );
739
740             // Play music-audio
741                 if( session[phase].music.content != "noMusic" ){
742                     $('#musicAudio').trigger('play');
743                 }
744
745             // Repeat random
746                 var musicPlayer = document.getElementById('musicAudio');
747
748                 musicPlayer.onended = function() {
749
750                     // Only in preview case
751                         if( !liveMode ){
752                             musicDelaySwitch = setTimeout(()=>{ startMusicPlayer(phase); },2000);
753                         }
754                     // Only in live-session
755                         else{
756                             var nextPhase = parseInt(phase) + 1;
757
758                             if( session[nextPhase] ){
759                                 if( phaseTimer > 5  ||  session[phase].music.content === session[nextPhase].music.content ){
760                                     startMusicPlayer( session[0].livePhase );
761                                 }
762                             }
763                             else{
764                                 if( phaseTimer > 5 ){
765                                     startMusicPlayer( session[0].livePhase );
766                                 }
767                             }
768                         }
769                 }
770         }
```

Nature-Player

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Aufruf zur Übergabe der src an das Audio-Element mit getAudioSrc().

Bereitstellen der src und Abspielen des Players mit dem Start der jeweiligen Phase: *startNaturePlayer(phase)*

Solange der Timer der Phase und Session noch nicht abgelaufen ist, wird die Audio in Endlosschleife überlappend wiedergegeben.

```
772     ////////////////////////////
773     // Nature
774     ////////////////////////////
775
776         function startNaturePlayer(phase){
777
778             const audio  = document.getElementById('natureAudio');
779
780             // As a precaution, stop it first if it was running
781                 audio.removeEventListener( 'timeupdate', startNaturePlayer );
782                 audio.currentTime = 0;
783                 audio.pause();
784
785             // Fill audio-player
786                 audio.src = getAudioSrc( eval(session[phase].nature.content) );
787
788            // Repeat random
789                 const overlapTime = 2;
790                 audio.addEventListener('timeupdate', () => {
791                     if (audio.currentTime >= audio.duration - overlapTime) {
792                       audio.currentTime = overlapTime;
793                     }
794                 });
795                 audio.play();
796         }
```

js/main.js

Sound-Player

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Aufruf zur Übergabe der src an das Audio-Element mit getAudioSrc().

Bereitstellen der src und Abspielen des Players mit dem Start der jeweiligen Phase:
startSoundPlayer(phase)

Konfigurationen:
1x am Anfang der Phase abspielen
1x am Ende der Phase abspielen
Am Anfang und Ende 1x abspielen
Im Interval abspielen

```
798     ////////////////////////////
799     // Sound
800     ////////////////////////////
801
802         var finalSound;
803
804         function startSoundPlayer(phase){
805             // As a precaution, stop it first if it was running
806                 $('#soundAudio').trigger('pause');
807                 clearInterval(finalSound);
808
809             // Fill audio-player
810                 $('#soundAudio').attr(  'src', getAudioSrc( eval(session[phase].sound.content) ) );
811                 $('#soundSource').attr( 'src', getAudioSrc( eval(session[phase].sound.content) ) );
812
813             // Interval-Cases
814                 if( session[phase].sound.interval == "start" ){
815                     $('#soundAudio').trigger('play');
816                 }
817                 else if( session[phase].sound.interval == "end" ){
818                     // Also start at beginning to preview user the sound (only soundscapes)
819                     if(session.length<=2){
820                         $('#soundAudio').trigger('play');
821                     }
822                     if(liveMode){
823                         finalSound = setInterval( function(){
824                                 if( phaseTimer <= 0 ){
825                                     $('#soundAudio').trigger('play');
826                                     clearInterval(finalSound);
827                                 }
828                         }, 1000);
829                     }
830                 }
831                 else if( session[phase].sound.interval != 0  &&  session[phase].sound.interval != "off" ){
832                     $('#soundAudio').trigger('play');
833                     // Not in preview, only live
834                         if(liveMode){
835                             var soundRepeater = setInterval( function(){
836                                 // Repeat sound in interval
837                                     phaseTimer % (session[phase].sound.interval*60) == 0   ?   $('#soundAudio').trigger('play')  :  null;
838                                 // Exit
839                                     phaseTimer <= 3  ?  clearInterval(soundRepeater)  :  null;
840                             }, 600);
841                         }
842                 }
843
844             // Repeat alarm if necessary
845                 var soundPlayer = document.getElementById('soundAudio');
846
847                 soundPlayer.onended = function() {
848                     if( session[0].timer > 5  &&  eval(session[phase].sound.content).category == "Wecker" ){
849                         startSoundPlayer(phase);
850                     }
851                 }
852
853         }
```

js/main.js

Noise-Player

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Kein HTML <audio>-Element.

Stattdessen Verwendung der Web-Audio-API zur weiteren Modulation der Audiodatei mit Splitter, Bandpass-Filter etc.

Statische Übergabe der src in
switch (session[phase].noise.content) {...}

Aufruf zur Übergabe der src und dem Starten des Players mit
startNoise(phase)

Abspielen in Endlosschleife

```
847  ////////////////////////////  NOISE  ////////////////////////////////////////////////////////////////////////////////////////////////////
848
849  var noiseFadeInFinished;
850  var noiseFadeOutStarted;
851
852  function stopNoise(){ ...
880  }
881
882  function muteNoise(){ ...
886  }
887
888  function unMuteNoise(){ ...
892  }
893
894  function startNoise(phase){
895
896      if( session[phase].noise.content != "noNoise"  &&  session[0].enableFreq ){
897
898              // Clear Intervals if stop
899                  destroySweepIntervals = false;
900
901            // Audio-Elements
902              if( !session[phase].noise.ctx ){ session[phase].noise.ctx = new AudioContext() }
903
904              let src;
905                  switch (session[phase].noise.content) {
906                      case "white":
907                      case "whiteSync":
908                      src = "audio/noise/white.mp3";
909                      break;
910                      case "pink":
911                      case "pinkSync":
912                      src = "audio/noise/pink.mp3";
913                      break;
914                      case "brown":
915                      case "brownSync":
916                      src = "audio/noise/brown.mp3";
917                      break;
918                      case "grey":
919                      case "greySync":
920                      src = "audio/noise/grey.mp3";
921                      break;
922                      case "blue":
923                      case "blueSync":
924                      src = "audio/noise/blue.mp3";
925                      break;
926                      case "violet":
927                      case "violetSync":
928                      src = "audio/noise/violet.mp3";
929                      break;
930              }
931
932              session[phase].noise.audio  = new Audio(src);
933              session[phase].noise.source = session[phase].noise.ctx.createMediaElementSource( session[phase].noise.audio );
934              session[phase].noise.audio.play();
935              session[phase].noise.audio.loop = true;
936
937            // Split Source L/R-Channel
938              session[phase].noise.splitter = session[phase].noise.ctx.createChannelSplitter(2);
939
```

js/generator-frequency-noise.js

Dateipfad getAudioSrc()

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Konfiguration der Session => Dateipfad
1) Auslesen des Strings aus der konfigurierten Session mit
session[phase].text.content
|| music.content || nature.content
|| noise.content || sound.content

2) Umwandlung des Strings in ein Objekt (jeweils hinterlegt in js/ content-database.js
eval("string") => DB-Objekt dbItem
3) Objekt dbItem macht z.B. Aussage über die Anzahl der Songs in einem Musikgenre, die Länge der Text-Audiodateien der verschiedenen Sprecher oder das Abspielverhalten wie "one" oder "loop"

Case: Nature / Noise / Sound
return "../path/audiofile.mp3";

Case: Text
Ein Dateipfad wird generiert und um den Namen des Sprechers ergänzt:
../path/voicename_textname.mp3

Case: Text (mehrere Dateien)
Wie im Case zuvor, doch wird hier der Dateiname um eine zufällige Nummer, je nach Anzahl der Verfügbaren Dateien generiert. Gleichzeitig wird dafür gesorgt, dass dieselbe Nummer frühestens dann nochmal generiert wird, wenn zuvor mind. 2/3 aller anderen verfügbaren Audiodateien abgespielt wurden.
../path/voicename_textname_45.mp3

Case: Musik (immer mehrere Dateien)
Wie im Case zuvor.
Bei Musik entfällt der Sprechername
../path/musikgenre_123.mp3

```
449
450 ////////////
```